THE MIRROR LAW (mLA): WHEN THE NORMALIZED MAGMOID IS THE QUOTIENT $\mathcal{A}/\ker K$, AND THE TWO CLASSICAL ROADS TO ASSOCIATIVITY

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ABSTRACT. In the positive home of the normalization defect, a *normalizer* K on an associative category $\mathcal{A}$ satisfies the left-absorption law (LA) $K(K(x)\cdot c) = K(x\cdot c)$ and generates a magmoid $\mathbf{Norm}_K = \operatorname{Im} K$ with product $f \# g := K(f\cdot g)$; this magmoid is associative exactly when the two-sided law ($\mathsf{RA}_\#$) holds, and *handedness* is (LA)$\wedge\neg$($\mathsf{RA}_\#$). Two visibly different families of normalizers are nonetheless associative — the normal-subgroup coset maps, and certain order-3 non-invertible monoids — and a referee asked for the single condition behind both. We answer it. Define the *mirror law* (mLA) $K(c\cdot K(x)) = K(c\cdot x)$, the two arguments of (LA) swapped. We prove three things. (1) For a normalizer, (mLA) alone forces ($\mathsf{RA}_\#$), and (LA)$\wedge$(mLA) makes the kernel relation $a \sim_K b :\Leftrightarrow K(a) = K(b)$ a *two-sided congruence*, so that $\mathbf{Norm}_K \cong \mathcal{A}/\!\sim_K$ as categories — the magmoid is literally $\mathcal{A}$ with $\sim_K$-equal arrows identified; indeed (mLA) is *equivalent* to "$\sim_K$ is a left congruence". (2) (mLA) is strictly stronger than ($\mathsf{RA}_\#$), and incomparable with closure of $\operatorname{Im} K$: each holds while the other fails, so they are two genuinely different sufficient conditions, forming a diamond over associativity. (3) For a group G with K the right-coset-representative map of $H \le G$, (mLA)$\Leftrightarrow H \triangleleft G$ (then $\mathbf{Norm}_K \cong G/H$); without normality, writing $g\cdot h = \varphi(g,h)\cdot(g\#h)$ with $\varphi \in H$, the transversal product is associative $\Leftrightarrow f\varphi(g,h)f^{-1} \in H$ for all f, g, h in the transversal. Closure ($\varphi \equiv 1$) and normality ((mLA)) are the two canonical solutions of this one conjugation criterion. The results are proved by hand, corroborated by exhaustive enumeration over all monoids of order ≤ 3 and all (H, T) transversal pairs of S_3 (15) and S_4 (35 416), zero mismatches, and the finite core is machine-checked axiom-free in Lean 4.

1. The question, and the setting

This note is a companion to the triangulation unit (*What normalization's non-associativity is, and is not*) and its closure criterion. We use that unit's vocabulary and recall only what the three theorems below consume. Composition is diagrammatic: $f \cdot g$ reads "f then g".

Definition 1.1 (normalizer, normalized magmoid)**.** A *normalizer* on an associative category $(\mathcal{A}, \cdot, \mathrm{id})$ is a family of endomaps $K_{X,Y} : \mathcal{A}(X,Y) \to \mathcal{A}(X,Y)$ satisfying, for all composable x, c,

- (K1) $K(Ka) = Ka$ (idempotent),
- (U) $K(\mathrm{id}_X) = \mathrm{id}_X$ (unital),
- (LA) $K\big(K(x)\cdot c\big) = K(x\cdot c)$ (left absorption).

Its *normalized magmoid* $\mathbf{Norm}_K$ has the same objects, hom-sets $\operatorname{Im} K = \{a : K(a) = a\}$ (equal by (K1)), product $f \# g := K(f\cdot g)$, and units id_X. We call $\operatorname{Im} K$ *closed* when $f, g \in \operatorname{Im} K \Rightarrow f\cdot g \in \operatorname{Im} K$, i.e. $K(f\cdot g) = f\cdot g$.

The magmoid $\mathbf{Norm}_K$ is a category-minus-associativity; the triangulation unit records (its Theorem B) exactly when associativity returns.

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[0]MacBeth (scotmacbeth) · 2026-10-23 (uid 242 corrections, 2026-10-24) · For: Rick · content commit `6df0043`. A self-contained write-up of the registry node `mirror-LA-sufficient-for-associativity` (proof `proofs/2026-10-22-mirror-LA-sufficient-for-associativity.md`), under the peer-reviewed positive-home parent. Answers SHOULD S1 of the uid 239 referee report on the closure criterion. The finite core is Lean-certified (`lean-mirror-LA-oracle`). This note recaps, and does not re-prove, Theorems A and B of the positive-home note.

Proposition 1.2 (Theorem B of the positive-home note, recalled)**.** $\mathbf{Norm}_K$ *is associative if and only if the two-sided law*

$$(\mathsf{RA}_{\#}): \qquad K\big(f \cdot K(g \cdot h)\big) = K(f \cdot g \cdot h) \qquad (f, g, h \in \operatorname{Im} K)$$

holds. The normalizer K *is* handed *when* (LA) *holds but* ($\mathsf{RA}_{\#}$) *fails; handedness is the non-associativity of* $\mathbf{Norm}_K$*.*

Two families of normalizers turn out to be associative for reasons that look nothing alike. The first: $\mathcal{A} = G$ a group, K the right-coset-representative map of a *normal* subgroup, where $\mathbf{Norm}_K \cong G/H$. The second: the small non-invertible monoids of the closure criterion's converse witness — the order-3 normalizers of the triangulation unit's Theorem 2(a) whose image is *not* closed, yet whose magmoid is associative (e.g. $M = \{e, a, a^2\}$, $a^3 = a$, with $K(a^2) = a$). These are associative for a reason unrelated to the group quotient, and (as §3 shows) *not* because the image is closed. The referee's question (uid 239, SHOULD S1) was whether one condition sits behind both. It does.

Definition 1.3 (kernel relation, mirror law)**.** On each hom-set, $a \sim_K b :\Leftrightarrow K(a) = K(b)$; write $[a]$ for the class and $\mathcal{A}/\!\sim_K$ for the quotient graph with hom-sets $\mathcal{A}(X, Y)/\!\sim_K$. The normalizer K satisfies the *mirror law* — (LA) with its two arguments swapped — if

$$(\mathsf{mLA}): \qquad K\big(c \cdot K(x)\big) = K(c \cdot x) \qquad \text{for all composable } c, x \in \mathcal{A}.$$

Remark 1.4 ((mLA) is a genuine constraint). As with (LA), the variable x ranges over *all* of $\mathcal{A}$, in particular over un-normalized products $x = g \cdot h$ of $\mathbf{Norm}_K$-arrows. Where $K(x) \neq x$, (mLA) asserts that the inner normalization is invisible to a *left* multiply-then-normalize. It is *not* the triviality $K(c \cdot K(x)) = K(c \cdot x)$ for $x \in \operatorname{Im} K$ (where both sides are $K(c \cdot x)$ by (K1)). This is the mirror of Remark 1.3 of the positive-home note.

Remark 1.5 (provenance). The mirror condition (mLA) and its core consequences were first stated by the referee (Rick, uid 239 report, SHOULD S1): that (mLA) yields ($\mathsf{RA}_{\#}$) immediately, that $\mathbf{Norm}_K \cong \mathcal{A}/\ker K$, that over a group the coset reading forces H normal, that it is strictly stronger than ($\mathsf{RA}_{\#}$), and the exact group criterion, with the enumeration counts. What this note contributes is the *written proofs* below; the converse of Remark 2.2 ((mLA) *equivalent* to "$\sim_K$ a left congruence," independently of (LA)); the closed-image separator of Theorem 3.1, which establishes that (mLA) and closure are genuinely *incomparable*; the core-of-H form of the group criterion (Remark 4.3); the honest-unifier Proposition 5.1; and the axiom-free Lean core. We flag the separation so the referee's prior claim is credited where it is due.

The three theorems say: (§2) (mLA) is exactly the missing left-handed half, and it upgrades associativity to a quotient; (§3) it is strictly stronger than associativity and incomparable with closure; (§4) over groups it *is* normality, and it names the exact boundary.

2. The mirror law makes $\mathbf{Norm}_K$ the quotient $\mathcal{A}/\ker K$

Theorem 2.1. *Let* K *be a normalizer satisfying* (mLA)*. Then:*

1. ($\mathsf{RA}_{\#}$) *holds; hence* $\mathbf{Norm}_K$ *is associative.*
2. $\sim_K$ *is a* two-sided congruence *on* $\mathcal{A}$*: if* $a \sim_K a'$ *and* $b \sim_K b'$ *(composable) then* $a \cdot b \sim_K a' \cdot b'$*. Consequently* $\mathcal{A}/\!\sim_K$ *is a category under* $[a] \cdot [b] := [a \cdot b]$*, units* $[\mathrm{id}_X]$*.*
3. $\Phi\colon \mathbf{Norm}_K \to \mathcal{A}/\!\sim_K$*,* $\Phi(f) = [f]$*, is an* isomorphism of categories*, with inverse* $\Psi([a]) = K(a)$*. In particular* $\mathbf{Norm}_K \cong \mathcal{A}/\!\sim_K$*.*

The key idea is that a normalizer always carries *one* half of a congruence — (LA) is the right-handed half — and (mLA) supplies the other. With both halves the kernel is a congruence, and the quotient of a category is a category, automatically associative; the normalize-then-compose product $\#$ is *forced* to agree with the quotient product because $K(f \cdot g) \sim_K f \cdot g$. That is the whole mechanism.

Proof. **(1).** Instantiate (mLA) at $c := f$ and $x := g \cdot h$, with $f, g, h \in \operatorname{Im} K$:

$$K\big(f \cdot K(g \cdot h)\big) = K\big(f \cdot (g \cdot h)\big) = K(f \cdot g \cdot h).$$

The left-hand side is the left-hand side of ($\mathsf{RA}_{\#}$); the right-hand side is its right-hand side. So ($\mathsf{RA}_{\#}$) holds, and $\mathbf{Norm}_K$ is associative by Proposition 1.2. (The derivation of ($\mathsf{RA}_{\#}$) here uses (mLA) only; the passage from ($\mathsf{RA}_{\#}$) to *associativity* is Proposition 1.2 — Theorem B of the positive-home note — whose proof uses (LA). So the hand argument for "(mLA) ⇒ associative" presupposes (LA); (LA) also enters directly in part (2).)

(2). We establish the two one-sided compatibilities separately.

Right-compatibility, from (LA). Read (LA) backwards as $K(x \cdot c) = K(K(x) \cdot c)$. If $K(a) = K(a')$ with $a, a' \in \mathcal{A}(X, Y)$ and $b \in \mathcal{A}(Y, Z)$, then

$$K(a \cdot b) = K\big(K(a) \cdot b\big) = K\big(K(a') \cdot b\big) = K(a' \cdot b),$$

so $a \cdot b \sim_K a' \cdot b$. This holds for *every* normalizer.

Left-compatibility, from (mLA). If $K(b) = K(b')$ with $b, b' \in \mathcal{A}(Y, Z)$ and $a \in \mathcal{A}(X, Y)$, then applying (mLA) with $c := a$,

$$K(a \cdot b) = K\big(a \cdot K(b)\big) = K\big(a \cdot K(b')\big) = K(a \cdot b'),$$

so $a \cdot b \sim_K a \cdot b'$.

Now if $a \sim_K a'$ and $b \sim_K b'$ then $K(a \cdot b) = K(a' \cdot b)$ (right-compatibility) $= K(a' \cdot b')$ (left-compatibility, common first factor a'), i.e. $a \cdot b \sim_K a' \cdot b'$. So $\sim_K$ is a two-sided congruence, and the quotient graph inherits a well-defined composition $[a] \cdot [b] := [a \cdot b]$, associative and unital because $\cdot$ is, hence a category.

(3). Ψ is well-defined: $a \sim_K a' \Rightarrow K(a) = K(a') \in \operatorname{Im} K = \mathbf{Norm}_K$. The two maps are mutually inverse:

$$\Psi(\Phi(f)) = \Psi([f]) = K(f) = f \quad (f \in \operatorname{Im} K), \qquad \Phi(\Psi([a])) = \Phi(K(a)) = [K(a)] = [a],$$

the last because $K(K(a)) = K(a)$ gives $K(a) \sim_K a$ by (K1). And Φ is functorial:

$$\Phi(f \mathbin{\#} g) = \big[K(f \cdot g)\big] = [f \cdot g] = [f] \cdot [g] = \Phi(f) \cdot \Phi(g), \qquad \Phi(\mathrm{id}_X) = [\mathrm{id}_X],$$

using $K(f \cdot g) \sim_K f \cdot g$ for the second equality. Being bijective on objects and on each hom-set, Φ is an isomorphism of categories. □

Remark 2.2 ((mLA) is exactly "$\sim_K$ a left congruence"). Theorem 2.1(2) shows (mLA) ⇒$\sim_K$ a left congruence. The converse holds using only (K1): if $\sim_K$ is a left congruence, then since $K(x) \sim_K x$, left-compatibility with first factor c gives $K(c \cdot K(x)) = K(c \cdot x)$, which is (mLA). Hence

$$\text{(mLA)} \iff \sim_K \text{ is a left congruence,}$$

independently of (LA); dually, (LA) ⇔$\sim_K$ is a *right* congruence. This is the reading Theorem 4.1 uses.

Remark 2.3 (what Theorem 2.1 adds to Theorem B). Proposition 1.2 says *when* $\mathbf{Norm}_K$ is associative; Theorem 2.1 says *what it then is*. ($\mathsf{RA}_{\#}$) is the bare ternary associativity equation; (mLA) is the stronger, more legible statement that the kernel is a congruence, so $\mathbf{Norm}_K$ is $\mathcal{A}$ with $\sim_K$-equal arrows identified. A normalizer always supplies the (LA) (right) half of the congruence, so (mLA) is precisely the missing left half. Both associative families of §1 will turn out to be quotients $\mathcal{A}/\!\sim_K$.

3. Strictly stronger than associativity, and incomparable with closure

(mLA) is a *sufficient* condition for associativity. So is closure of $\operatorname{Im} K$ (the triangulation unit's closure criterion). Neither is necessary, and — the point of this section — neither implies the other.

Theorem 3.1. *There is a normalizer with* $(\mathsf{LA}) \wedge (\mathsf{RA}_{\#}) \wedge \neg(\mathsf{mLA})$*; so* $(\mathsf{mLA}) \Rightarrow (\mathsf{RA}_{\#})$ *is a proper implication. Moreover* (mLA) *and closure of* $\operatorname{Im} K$ *are* incomparable*: each can hold while the other fails. They are two distinct sufficient conditions, both strictly above associativity.*

Proof. A separator with $(\mathsf{RA}_{\#}) \wedge \neg(\mathsf{mLA})$ *and* $\operatorname{Im} K$ *closed.* Let $M = \{0, 1, 2\}$ with 0 the unit and multiplication (rows = left factor)

$\cdot$	0	1	2
0	0	1	2
1	1	1	2
2	2	1	2

M is associative and $1, 2$ are non-invertible (both checked exactly). Take $K(0) = 0,\ K(1) = 0,\ K(2) = 2$, so $\operatorname{Im} K = \{0, 2\}$. Then $(\mathsf{K1})$, (U), (LA) hold (checked over all 3×3 pairs), and $\operatorname{Im} K$ is *closed* ($0 \cdot 0 = 0$, $0 \cdot 2 = 2$, $2 \cdot 0 = 2$, $2 \cdot 2 = 2$), so $(\mathsf{RA}_{\#})$ holds and $\mathbf{Norm}_K$ — the two-element monoid $\{0, 2\}$ with 2 absorbing — is associative. But (mLA) fails at $c = 2,\ x = 1$:

$$K\big(2 \cdot K(1)\big) = K(2 \cdot 0) = K(2) = 2, \qquad K(2 \cdot 1) = K(1) = 0, \qquad 2 \neq 0.$$

Thus $(\mathsf{RA}_{\#}) \wedge \neg(\mathsf{mLA})$ — so $(\mathsf{mLA}) \Rightarrow (\mathsf{RA}_{\#})$ is proper — and, since $\operatorname{Im} K$ is closed here, *closure does not imply* (mLA).

A separator with (mLA) *but* $\operatorname{Im} K$ *not closed.* The order-3 converse witness of the closure note ($M = \{e, a, b\}$ with $a \cdot a = b$, $a \cdot b = b \cdot a = a$, $b \cdot b = b$ — i.e. $M = \{e, a, a^2\}$ with $a^3 = a$ — and $K(e) = e$, $K(a) = a$, $K(b) = a$) has $\sim_K$-classes $\{e\}, \{a, b\}$, a two-sided congruence, so (mLA) holds; yet $\operatorname{Im} K = \{e, a\}$ is not closed because $a \cdot a = b \notin \operatorname{Im} K$. So *$(\mathsf{mLA})$ does not imply closure.* The two conditions are incomparable. □

The failure in the first separator is diagnostic. Closure says the *image* is a subcategory; (mLA) says the *kernel* $\sim_K$ is a left congruence. Here the classes are $\{0, 1\}$ and $\{2\}$, and left-multiplying the class $\{0, 1\}$ by 2 sends $2 \cdot 0 = 2$ into class $\{2\}$ but $2 \cdot 1 = 1$ into class $\{0, 1\}$: left multiplication by 2 does not respect $\sim_K$. A closed image is blind to this; (mLA) is not.

Remark 3.2 (the diamond). Theorem 3.1 draws a diamond over associativity:

$$\mathbf{Norm}_K \text{ associative } = (\mathsf{RA}_{\#})$$

$$\nearrow \qquad\qquad\qquad\qquad \nwarrow$$

$(\mathsf{mLA}) : \sim_K$ a left congruence — [(mLA) holds, not closed]

closure: $\operatorname{Im} K$ a subcategory — [closed, (mLA) fails]

Each upward arrow is strict and the two upper conditions are logically independent. Over groups (§4) they are the two classical mechanisms: $(\mathsf{mLA}) \Leftrightarrow H$ normal, closure $\Leftrightarrow T$ a subgroup. Handedness $= (\mathsf{LA}) \wedge \neg(\mathsf{RA}_{\#})$ therefore requires both $\neg(\mathsf{mLA})$ and ¬closure — each is necessary for associativity to fail, neither alone is the obstruction.

One caution: the bottom vertex is *not* the join of the two upper ones. Associativity ($(\mathsf{RA}_{\#})$) is strictly weaker than "(mLA) or closure": there are normalizers that are associative with *neither* (mLA) nor closed image. The smallest such lie at order 4 among monoids (six normalizers, found by exhaustive search; §6) and at $S_3 \times C_2$ among groups. Only at order ≤ 3 do the two coincide — there associativity $= (\mathsf{mLA}) \vee$ closure exactly, with zero exceptions — which is why the two-condition picture looks complete on the small examples but is not the general story. The general account is Proposition 5.1.

4. Over a group: the mirror law is normality

Let G be a group, $H \leq G$ a subgroup, and T a *right transversal* (one representative per right coset Hg, with $\mathrm{id} \in T$). Let $K\colon G \to G$ send g to the unique $t \in T$ with $Hg = Ht$; then $\operatorname{Im} K = T$ and $\#$ is the transversal (right-loop) product $t \,\#\, t' = K(t \cdot t')$.

Two facts are immediate. First, (LA) holds for *every* transversal rep map: since $K(x) \in Hx$, we have $H \cdot K(x) = H \cdot x$, hence $H \cdot (K(x) \cdot c) = H \cdot (x \cdot c)$ and $K(K(x) \cdot c) = K(x \cdot c)$. Second, $K(a) = K(b) \Leftrightarrow Ha = Hb$, so the classes of $\sim_K$ are exactly the *right cosets* of H.

Theorem 4.1. *With K the right-coset-representative map of $H \leq G$: (mLA) holds $\Leftrightarrow H \triangleleft G$. When $H \triangleleft G$, Theorem 2.1(3) specializes to the classical isomorphism $(T, \#) \cong G/H$.*

Proof. By Remark 2.2, (mLA) $\Leftrightarrow \sim_K$ is a left congruence. The $\sim_K$-classes are right cosets, so left-compatibility reads: $Ha = Hb \Rightarrow H(ca) = H(cb)$ for all c. Equivalently, for all a, b, c: $ab^{-1} \in H \Rightarrow (ca)(cb)^{-1} = c(ab^{-1})c^{-1} \in H$. Letting ab^{-1} range over all of H and c over G, this is exactly $cHc^{-1} \subseteq H$ for all c, i.e. $H \triangleleft G$. When H is normal, $\sim_K$ is equality of (two-sided) cosets, so $\mathcal{A}/\sim_K = G/H$ and $\Phi\colon t \mapsto Ht$ is the group isomorphism $(T, \#) \cong G/H$. □

So the normal-subgroup family of associative transversals *is* the (mLA) family. The remaining associative transversals — associative with H not normal — are governed by an explicit ternary criterion.

Theorem 4.2 (the exact boundary)**.** *For $f, g, h \in T$ write the de-normalization $g \cdot h = \varphi(g, h) \cdot (g \# h)$ with $\varphi(g, h) := (g \cdot h)(g \# h)^{-1} \in H$. Then $(T, \#)$ is associative $\Leftrightarrow f\,\varphi(g, h)\,f^{-1} \in H$ for all $f, g, h \in T$.*

Proof. Since $g \# h = K(g \cdot h) \in Hg \cdot h$, indeed $\varphi(g, h) \in H$ and $g \# h = \varphi(g, h)^{-1} \cdot (g \cdot h)$. By (LA) and Theorem A of the positive-home note, $(f \# g) \# h = K(f \cdot g \cdot h)$; and

$$f \# (g \# h) = K\big(f \cdot (g \# h)\big) = K\big(f \cdot \varphi(g, h)^{-1} \cdot g \cdot h\big).$$

Associativity is $K(f \cdot g \cdot h) = K(f \cdot \varphi(g, h)^{-1} \cdot g \cdot h)$ for all $f, g, h \in T$. Using $K(x) = K(y) \Leftrightarrow xy^{-1} \in H$, this is

$$(f \cdot g \cdot h)\big(f \cdot \varphi(g, h)^{-1} \cdot g \cdot h\big)^{-1} = f \cdot g \cdot h \cdot (g \cdot h)^{-1} \cdot \varphi(g, h) \cdot f^{-1} = f\,\varphi(g, h)\,f^{-1} \in H$$

for all $f, g, h \in T$. □

Remark 4.3 (core form: the f-quantifier disappears). Because $G = HT$ and $(hf)^{-1}H(hf) = f^{-1}Hf$ for $h \in H$, running f over the transversal T is the same as running it over all of G, so $\bigcap_{f \in T} f^{-1}Hf = \bigcap_{g \in G} g^{-1}Hg = \mathrm{core}_G(H)$, the *normal core* — the largest normal subgroup of G contained in H. Thus "$f\varphi(g, h)f^{-1} \in H$ for all f" is just $\varphi(g, h) \in \mathrm{core}_G(H)$, and Theorem 4.2 becomes

$$(T, \#) \text{ associative} \iff \varphi(T \times T) \subseteq \mathrm{core}_G(H) \iff \text{the image of } T \text{ in } G/\mathrm{core}_G(H) \text{ is a subgroup.}$$

The two ends of the diamond are the two extremes of the core: (mLA) is $\mathrm{core}_G(H) = H$ (i.e. $H \triangleleft G$), and closure is $\varphi \equiv \mathrm{id}$, where the core plays no role. This reads off the enumeration. In S_3 every proper subgroup is either normal (A_3) or has *trivial* core (the order-2 subgroups), and trivial core forces $\varphi \equiv \mathrm{id}$ on any associative transversal — i.e. closure — so S_3 has *no* strictly-between transversal; the first groups with a strictly-between regime are those with a non-trivial proper core, e.g. $S_3 \times C_2$ (core C_2), which has nine.

Remark 4.4 (the criterion is the diamond). The criterion $f\varphi(g, h)f^{-1} \in H$ is satisfied in two canonical ways, matching Remark 3.2:

- *closure / T a subgroup:* then $g \cdot h \in T$, so $g \# h = g \cdot h$ and $\varphi(g, h) = \mathrm{id}$; the criterion is vacuous.
- *normality / (mLA):* $\varphi(g, h) \in H$ and $H \triangleleft G$, so $f\varphi(g, h)f^{-1} \in H$ automatically.

The criterion is strictly weaker than either: it permits a nontrivial φ, valued in a non-normal H, that merely stays in the normal core $\mathrm{core}_G(H)$ (Remark 4.3). That strictly-between regime is the honest content of "associative without normality or closure".

5. The honest unifier: one condition, two specializations

Sections 3–4 might read as though, over a *general* associative category $\mathcal{A}$, the two sufficient conditions — (mLA) and closure of $\operatorname{Im} K$ — were the two faces of one mechanism. They are not. Theorem 3.1 shows they are incomparable, and off the group setting there is no φ to carry a common criterion: "one mechanism and its two classical degenerations" is a group phenomenon (§4), not a general theorem. What *is* general is the following, isolated by the referee (uid 239) and adopted here as the honest statement of "one condition behind both". It pins down the exact restriction of (mLA) that is equivalent to associativity.

Write θ_K for the *largest two-sided congruence contained in* $\sim_K$; explicitly,

$$a\,\theta_K\,b \;:\Longleftrightarrow\; K(x\cdot a\cdot y)=K(x\cdot b\cdot y)\quad\text{for all composable } x,y.$$

(Taking $x=y=\mathrm{id}$ shows $\theta_K\subseteq{\sim_K}$; the two quantifiers make θ_K a two-sided congruence; and any two-sided congruence inside $\sim_K$ is contained in θ_K, so it is the largest.)

Proposition 5.1 (the honest unifier)**.** *Let K be a normalizer (so (LA) holds). The following are equivalent.*

1. $\mathbf{Norm}_K$ *is associative (equivalently, (RA$_\#$) holds).*
2. *(mLA) holds* on products of normalized arrows: $K\big(c\cdot K(x)\big)=K(c\cdot x)$ *for all $c\in\mathcal{A}$ and all $x\in\operatorname{Im}K\cdot\operatorname{Im}K$ (i.e. $x=g\cdot h$, $g,h\in\operatorname{Im}K$).*
3. $K(g\cdot h)\,\theta_K\,(g\cdot h)$ *for all $g,h\in\operatorname{Im}K$.*

Proof. (2)⇒(1). Take $c=f\in\operatorname{Im}K$ and $x=g\cdot h$ with $g,h\in\operatorname{Im}K$; then (2) reads $K\big(f\cdot K(g\cdot h)\big)=K(f\cdot g\cdot h)$, which is (RA$_\#$). So $\mathbf{Norm}_K$ is associative by Proposition 1.2.

(1)⇒(2). For $c\in\mathcal{A}$ and $g,h\in\operatorname{Im}K$,

$$K\big(c\cdot K(g\cdot h)\big)\overset{(\mathsf{LA})}{=}K\big(K(c)\cdot K(g\cdot h)\big)\overset{(\mathsf{RA}_\#)}{=}K\big(K(c)\cdot g\cdot h\big)\overset{(\mathsf{LA})}{=}K(c\cdot g\cdot h),$$

the outer steps being (LA) (read backwards and forwards) and the middle step (RA$_\#$) applied to $K(c),g,h\in\operatorname{Im}K$, available because (1)⇔(RA$_\#$).

(2)⇔(3). Since K is idempotent, $K(g\cdot h)\sim_K(g\cdot h)$ always; and every normalizer has the (LA) (right-congruence) half, i.e. $K(a\cdot y)=K(K(a)\cdot y)$. Condition (3) unfolds to $K\big(x\cdot K(g\cdot h)\cdot y\big)=K\big(x\cdot(g\cdot h)\cdot y\big)$ for all x,y. Setting $y=\mathrm{id}$ gives (2) with $c=x$. Conversely, given (2), for all x,y,

$$K\big(x\cdot K(g\cdot h)\cdot y\big)=K\big(K(x\cdot K(g\cdot h))\cdot y\big)=K\big(K(x\cdot g\cdot h)\cdot y\big)=K\big(x\cdot(g\cdot h)\cdot y\big),$$

using right-compatibility at the outer equalities and (2) (with $c=x$) at the middle. So (3) holds. □

Remark 5.2 (what (2) is, and what it is not). Condition (2) is the honest "single condition behind both": it is (RA$_\#$) rewritten as (mLA) restricted to products of normalized arrows. The full mirror law (mLA) (all $x\in\mathcal{A}$) and closure of $\operatorname{Im}K$ are its two *sufficient specializations* — full (mLA) gives (2) a fortiori; closure makes (2) *vacuous* (then $x=g\cdot h\in\operatorname{Im}K$, so $K(x)=x$ and both sides are $K(c\cdot x)$). They are *not* two degenerations of one criterion over general $\mathcal{A}$ (Theorem 3.1: incomparable), but two sufficient routes to the *same* necessary-and-sufficient condition (2)/(3). Over a group with K the coset-representative map, θ_K is the coset relation of $\mathrm{core}_G(H)$ and (3) is Theorem 4.2 in the core form of Remark 4.3. The equivalence (1)⇔(3) was checked exhaustively on all 1578 normalizers of all 156 order-4 monoids (unit fixed) and all monoids of order ≤ 3, with zero mismatches.

6. Verification and scope

Lean (finite core, axiom-free). The leaf `lean-mirror-LA-oracle` (`lean/2026-10-22-mirror-LA-oracle.lean`, Lean 4, Mathlib-free) machine-checks the decidable content: on the order-4 handed model, (mLA) fails (as it must, being handed); (mLA) forces (RA$_\#$) and #-associativity over *all*

monoids of order ≤ 3 and all idempotent unit-fixing K[1] (zero counterexamples); (mLA) $\Leftrightarrow\sim_K$ a left congruence over the same range (zero mismatches, Remark 2.2); the strictness separator exists at order 3 and not below; and the explicit separator of §3 (closed image, (RA$_\#$) true, (mLA) false at $c = 2, x = 1$) is certified. The bundled theorem reports *no* axiom dependence — not even `propext`; every `decide` reduces to `rfl`. We cite it rather than reproduce it.

Exhaustive enumeration. `scratch/2026-10-22-mirror-LA-verify.py` (exact integer / permutation arithmetic). Over all monoids of order ≤ 3 and all their normalizers: (mLA) $\Rightarrow$ (RA$_\#$), (mLA) $\Rightarrow$ associative, and (mLA) $\Leftrightarrow\sim_K$ left-congruence hold with zero exceptions; right-congruence never fails under (LA); and, with the unit fixed (11 labelled order-3 tables), exactly 2 separators (RA$_\#$) $\wedge\neg$(mLA) occur, both at order 3 (none at order ≤ 2); allowing any element to be the unit (33 tables) the count is 6. Over all (H, T) transversal pairs of S_3 (15) and S_4 (35 416): (LA) never fails, [(mLA) $\Leftrightarrow$ H normal] has zero mismatches, and [$(T, \#)$ associative $\Leftrightarrow$ the criterion] has zero mismatches. The explicit S_3 witness of the "closed but (mLA) fails" corner is $H = \langle(1\,2)\rangle$ (order 2, non-normal), $T = A_3$ (a subgroup, hence closed): (LA) ✓, (mLA) fails, $(T, \#) \cong C_3$ associative.

Honest scope. Theorems 2.1–4.2 are proved by hand (elementary: (K1), (LA), (mLA), associativity of $\cdot$, and, for §4, the right-coset structure); the enumerations corroborate but are not the proof. Minimality of the order-3 separator is a computational fact (exhaustive over order ≤ 3), as in the closure note, and nothing downstream depends on it. The group reading assumes a right transversal with id $\in T$; the left-transversal reading swaps (LA) $\leftrightarrow$ (mLA) and gives the mirror statements. The group specialization (Theorem 4.1) and the S_3/S_4 enumerations are Python-only (permutation searches, not reducible kernel `decide`s at that scale); the Lean leaf certifies the abstract finite core.

What this settles

There is one honest condition behind the two associative families, and it is *not* a single mechanism with two degenerations — over a general $\mathcal{A}$, (mLA) and closure are incomparable (Theorem 3.1). The common condition is associativity itself, read as the restricted mirror law of Proposition 5.1: $K(c \cdot K(x)) = K(c \cdot x)$ for x a product of normalized arrows, equivalently $K(g \cdot h)\,\theta_K\,(g \cdot h)$. Full (mLA) and closure of $\operatorname{Im} K$ are its two *sufficient specializations* — the first makes the kernel a two-sided congruence, so $\mathbf{Norm}_K \cong \mathcal{A}/\sim_K$ is the honest quotient; the second makes the restricted law vacuous. (mLA) is itself a legible, checkable sufficient condition ("$\sim_K$ is a left congruence"), strictly stronger than bare (RA$_\#$) and incomparable with closure; the two sit in a diamond over associativity whose bottom vertex is strictly below their join. Over groups, (mLA) is normality and closure is subgroup-hood, and both solve the single conjugation criterion $f\varphi(g, h)f^{-1} \in H$ — equivalently $\varphi(T \times T) \subseteq \operatorname{core}_G(H)$ (Remark 4.3) — which exhibits, between them, the genuine associative-without-either regime. This is the content SHOULD S1 asked for.

References

[1] MacBeth, *The positive home of the normalization defect: a one-sided-absorbing idempotent normalizer* (program note, 2026-10-19; registry node `positive-home-one-sided-absorbing-idempotent`). Source of Theorems A and B and the handedness definition recalled here.

[2] MacBeth, *What normalization's non-associativity is, and is not: three obstruction theories that miss it, and the idempotent normalizer that receives it* (triangulation unit, `expository/2026-10-21-normalization-defect-triangulation`, content commit `6df0043`). Contains the closure criterion and the provenance discussion (Baer; Lal; Kiechle; Grabowski).

[3] E. Di Lavore, M. Román, P. Székely, *Normalization is a sesquilaw*, arXiv:2510.01131. Origin of the magmoid **Norm** and the renormalization idempotent.

[1] This Lean range includes idempotent unit-fixing K that do *not* satisfy (LA), which the hand proof of Theorem 2.1(1) does not cover (its passage (RA$_\#$)$\Rightarrow$ associative uses (LA), via Proposition 1.2). The certified implication "(mLA)$\Rightarrow$ #-associative" is nonetheless true on the whole range: at order 3 there are exactly 6 such K with (mLA) but without (LA), and all 6 are associative. For those the statement is a verified computational fact, not a consequence of the hand argument.

[4] R. Baer, Nets and groups, *Trans. Amer. Math. Soc.* **46** (1939) 110–141.
[5] R. Lal, Transversals in groups, *J. Algebra* **181** (1996) 70–81.
[6] H. Kiechle, *Theory of K-Loops*, Springer LNM 1778 (2002).

The oidified (categorical) version of this transversal/right-loop construction is published (Grabowski, *An introduction to loopoids*): see the provenance discussion of the triangulation unit [2], whose source index logs it. It is not re-cited here directly pending a deep-read registration.

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